Lecture 16

04 March 2026 15:57

Recap $\min_{x \in S} f(x)$

Lemma: (a) If $S \subseteq \mathbb{R}^n$ is convex, $f: S \to \mathbb{R}$ is continuously differentiable and $x^* \in S$ is a local optimum then $\nabla f(x)^T(x - x^*) \geq 0$ for all $x \in S$. (necessary condition)

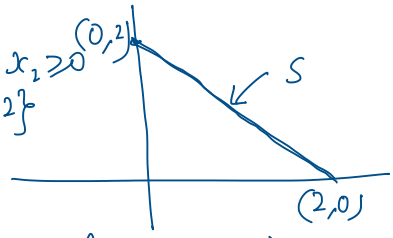
(b) In addition to the above assumptions, if f is also convex then $\nabla f(x)^T(x - x^*) \geq 0$ for all $x \in S$ is also a sufficient condition for x^* to be an optimum.

Example (Minimization over a simplex)

Simplex in $\mathbb{R}^n$ $S = \{x \in \mathbb{R}^n : x_i \geq 0 \ \forall i \text{ and } \sum_{i=1}^{n} x_i = r\}$

$\mathbb{R}^2$

Ex $S = \{x \in \mathbb{R}^2 : x_1 \geq 0, x_2 \geq 0, x_1 + x_2 = 2\}$



$\min_{x \in S} f(x)$ where $S = \{x \in \mathbb{R}^n : x_i \geq 0 \ \forall i, \sum x_i = r\}$

Suppose x^* is an optimal soln.

nec cond. $\nabla f(x^*)^T (x - x^*) \geq 0 \quad \forall x \in S$

Suppose $x_i^* > 0$

$$\left[\frac{\partial f(x^*)}{\partial x_1}, \dots, \frac{\partial f(x^*)}{\partial x_n}\right] \begin{bmatrix} 0 \\ 0 \\ \epsilon \\ \vdots \\ -\epsilon \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} \\ \\ \leftarrow j^{th} \text{ pos.} \geq 0, \ j \neq i \\ \\ \leftarrow i^{th} \text{ pos} \\ \\ \\ \end{matrix}$$

$x^* = \begin{bmatrix} x_1^* \\ x_2^* \\ \vdots \\ x_i^* \\ \\ x_n^* \end{bmatrix} \in S$ $\quad x = \begin{bmatrix} x_1^* \\ x_2^* \\ x_j^* + \epsilon \\ x_i^* - \epsilon \\ \\ x_n^* \end{bmatrix} \in S$ for small ϵ

$$\Leftrightarrow \epsilon\left(\frac{\partial f(x^*)}{\partial x_j} - \frac{\partial f(x^*)}{\partial x_i}\right) \geq 0$$

$$\Rightarrow \frac{\partial f(x^*)}{\partial x_i} \leq \frac{\partial f(x^*)}{\partial x_j} \quad \text{— (A)}$$

Suppose $x_j^* > 0$ then

$$\frac{\partial f(x^*)}{\partial x_j} \leq \frac{\partial f(x^*)}{\partial x_i} \quad \text{— (B)}$$

If $x_i^* > 0$, $x_j^* > 0$ then from (A), (B)

$$\frac{\partial f(x^*)}{\partial x_i} = \frac{\partial f(x^*)}{\partial x_j}$$

Example $\min -x_1 x_2 (x_1 - x_2)$

s.t. $\left\{\begin{matrix} x_1 + x_2 = 8 \\ x_1, x_2 \geq 0 \end{matrix}\right\}$ = a simplex in $\mathbb{R}^2$

if $x_1 = 0 \Rightarrow$ value $= 0$

$x_2 = 0 \Rightarrow$ value $= 0$

$\rightarrow$ value < 0

$x_1 > x_2 \Rightarrow$

Hence at optimality $x_1, x_2 > 0$ and $x_1 > x_2$

If x^* is an optimal soln $\Rightarrow x_1^* > 0,\ x_2^* > 0$ and $x_1^* > x_2^*$

$$\frac{\partial f(x^*)}{\partial x_1} = -2x_1^* x_2^* + x_2^{*2}$$

$$\frac{\partial f(x^*)}{\partial x_2} = -x_1^{*2} + 2x_1^* x_2^*$$

Because $x_1^* > 0,\ x_2^* > 0$ $\quad -2x_1^* x_2^* + x_2^{*2} = -x_1^{*2} + 2x_1^* x_2^*$

$$x_1^2 + x_2^2 - 4x_1x_2 = 0$$

$$\Leftrightarrow x_1^2 + 4x_2^2 - 4x_1x_2 - 3x_2^2 = 0$$

$$\Leftrightarrow (x_1 - 2x_2)^2 = (\sqrt{3}\,x_2)^2$$

$$\Leftrightarrow x_1 - 2x_2 = \pm\sqrt{3}\,x_2$$

$$\Leftrightarrow x_1 = (2 \pm \sqrt{3})\,x_2$$

$2+\sqrt{3} > 1$

$2-\sqrt{3} < 1$

On solving

$$x_1^* = (2+\sqrt{3})\,x_2^*$$

$$x_1^* + x_2^* = 8$$

$$(3+\sqrt{3})\,x_2^* = 8$$

$$x_2^* = \frac{8}{2+\sqrt{3}}$$

$$x_1^* = \frac{(2+\sqrt{3})\,8}{2+\sqrt{3}}$$

$$x^* = \begin{bmatrix} \frac{(2+\sqrt{3})8}{2+\sqrt{3}} \\ \frac{8}{3+\sqrt{3}} \end{bmatrix}$$

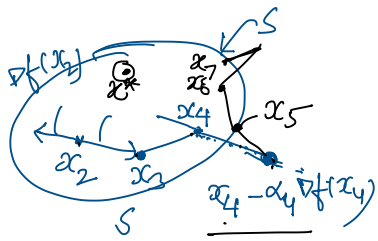
(this is the only point that satisfies the first order condition)

$$\nabla^2 f(x) = \begin{bmatrix} -2x_2 & -2x_1 + 2x_2 \\ -2x_1 + 2x_2 & 2x_1 \end{bmatrix}$$

(Need to analyze this to confirm whether x^* is optimal)

$$\min_{x \in S} f(x)$$

Iterative methods $\quad x_{t+1} = x_t - \alpha_t \nabla f(x_t)$ (unconstrained case)



x_5 is the closest point in S to $x_4 - \alpha_4 \nabla f(x_4)$

Now

$$x_{t+1} = [x_t - \alpha_t \nabla f(x_t)]^+$$

Given a set $S \subseteq \mathbb{R}^n$, a point x

$\underset{y \in S}{\arg\min} \|y - x\|$ is called Projection of x on S, denoted by $[x]^+$ $([x]_S)$

Suppose $x \in S \Rightarrow [x]^+ = x$

Projection: projection of x on S ... convex fn

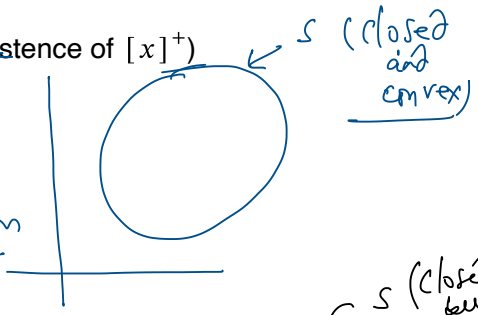
$$[x]^+ \triangleq \underset{y \in S}{argmin} \|x - y\|$$ = strictly

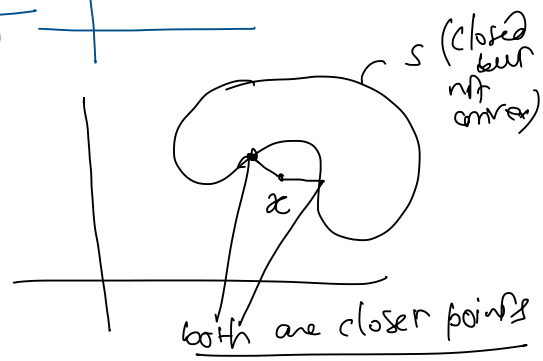
(a convex optimization problem, S being closed ~~ensures existence~~ of $[x]^+$)

$\min_{y \in S} \|y - x\|$

(is a convex optimization prob. in this case)

$\Rightarrow$ unique soln





Suppose S is convex

$[x]^+ = \arg\min_{y \in S} \|y - x\|$

$= \arg\min_{y \in S} \|y - x\|^2$

$\nabla_x (\|x\|^2) = 2x$

Necessary as well as suff. condition

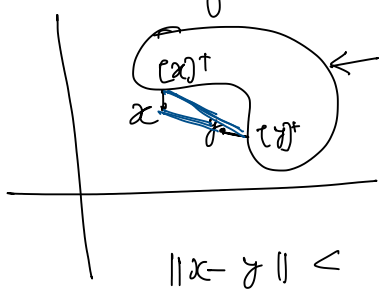
$2 (y - [x]^+)^T ([x]^+ - x) \geq 0$

$\min_{y \in S} f(x)$

$(y - x^*)^T \nabla f(x^*) \geq 0$

$\Leftrightarrow (y - [x]^+)^T (x - [x]^+) \leq 0$

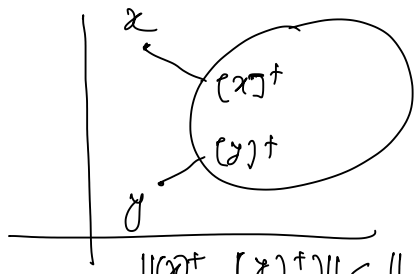
(necessary as well as suff.)



$\|x - y\| < \|[x]^+ - [y]^+\|$

Suppose S is convex

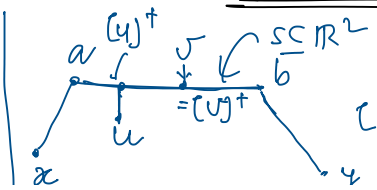
Ex. 1



$\|[x]^+ - [y]^+\| \leq \|x - y\|$

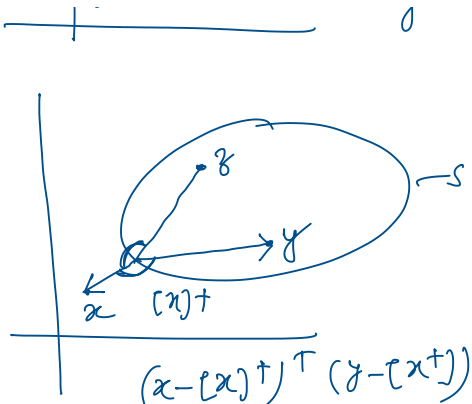
(nonexpansive)

Ex 2



$[x]^+ = a$

$[y]^+ =$

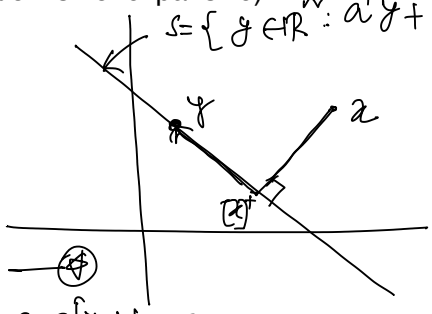


$(x-[x]^+)^T(y-[x^+]) \le 0$

Projection theorem: If $S \subseteq \mathbb{R}^n$ closed and convex

a. There exists a unique $[x]^+$ for each $x \in \mathbb{R}^n$,
b. For all $x \in \mathbb{R}^n$, $[x]^+$ is the projection iff $(y-[x]^+)^T(x-[x]^+) \le 0$ for all $y \in S$ (first order optimality condition)
c. $\|[x]^+ - [y]^+\| \le \|x-y\|$ for all $x, y \in \mathbb{R}^n$ (projection is nonexpansive)

Example



$\min \|y-x\|^2$

s.t. $a^T y + b = 0$ $\quad (y \in S$

first order cond.

$(y-[x]^+)^T(x-[x]^+) \ge 0$ — (*)

$\forall\, y \in a^T y + b = 0$

Any $y \in S$ can be written as $y = [x]^+ + z$

$\Rightarrow a^T z = 0$

$a^T[x]^+ + b = 0$

$a^T([x]^+ + z) + b = 0$

$\Rightarrow a^T z = 0$

$S = \{ [x]^+ + z : a^T z = 0 \}$

Now apply (*) $\quad z^T(x-[x]^+) \ge 0$ — (A) $\quad \forall z$ s.t. $a^T z = 0$

$\Rightarrow (-z)^T(x-[x]^+) \ge 0$ $\quad a^T(-z) = 0$

$\Rightarrow z^T(x-[x]^+) \le 0$ — (B)

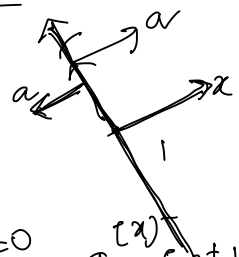
From (A) and B

$z^T(x-[x]^+) = 0 \quad \forall z : a^T z = 0$

this can happen only if

$x-[x]^+$ is aligned with a

i.e. $x-[x]^+ = \alpha a$ for some α



$(z^T a = 0 \Rightarrow z^T(x-[x]^+) = 0)$

$\Rightarrow [x]^+ = x - \alpha a$ $\quad (\alpha = ?)$

$[x]^+ \in S \Rightarrow a^T(x - \alpha a) + b = 0$

$\Rightarrow \alpha = \dfrac{a^T x + b}{a^T a} = \dfrac{a^T x + b}{\|a\|^2}$

$\Rightarrow [x]^+ = x - \dfrac{a^T x + b}{\|a\|^2} a$ $\quad x_{t+1} = (x_t - \alpha_t \nabla f(x_t)) - \dfrac{a^T(x_t - \alpha_t \nabla f(x_t)) + b}{\|a\|^2} \cdot a$

Gradient projection algorithm (steepest decent direction and fixed step-size rule):
Initialize: $x_0 \in \mathbb{R}^n$
Iterate: for $t \geq 0$, $x_{t+1} = [x_t - \alpha \nabla f(x_t)]^+$ where $\alpha > 0$

Recall Gradient descent

f is L-smooth $\quad \alpha \in (0, 2/L)$ then

$x_t \to x^*$ s.t. $\nabla f(x^*) = 0$ $\quad$ ($\bar{x}$ is fixed point of f $\Leftrightarrow f(\bar{x}) = \bar{x}$)

i.e., $x^* - \nabla f(x^*) = x^*$

i.e., x^* is a fixed point of $x - \nabla f(x)$

or, x^* is soln to $x - \nabla f(x) = x$

x^* is called a stationary point

Lemma: Let $S \subseteq \mathbb{R}^n$ be convex and $f: S \to \mathbb{R}$ be L-smooth. If $\alpha \in \left(0, \frac{2}{L}\right)$ then gradient projection algorithm converges to a $x^* \in S$ such that $x^* = [x^* - \alpha \nabla f(x^*)]^+$. ($x^*$ is a fixed point of $[x - \alpha \nabla f(x)]^+$)
(such a x^* is called a stationary point)

$\nabla f(x^*) = 0$ need not be true

But $(x - x^*)^T \nabla f(x^*) \geq 0$ $\quad \forall x \in S$

- If ~~f is convex~~ and Lipschitz continuous with parameter B then using ~~$\alpha = \dfrac{\|x_0 - x^*\|}{B\sqrt{T}}$~~,

$$\frac{1}{T}\sum_{t=0}^{T-1} (f(x_t) - f(x^*)) \leq \frac{\|x_0 - x^*\| B}{\sqrt{T}}$$ (lib. of mirror descent with para.)

- If f is convex and L-smooth then using $\alpha = \dfrac{1}{L}$,

$$f(x_T) - f(x^*) \leq \frac{L}{2T}\|x_0 - x^*\|^2$$

- If f is L-smooth and strongly convex with parameter m then using $\alpha = \dfrac{1}{L}$,

$$\|x_T - x^*\|^2 \leq \left(1 - \frac{m}{L}\right)^T \|x_0 - x^*\|^2$$

(these are exactly same as for gradient descent with fixed step size)

Constrained Problems with equality constraints

min $f(x)$

min $f(x)$

s.t. $h_1(x)=0$, $h_2(x)=0$, $\vdots$, $h_m(x)=0$ } m equality constraints

$h_i : \mathbb{R}^n \longrightarrow \mathbb{R}$ $\forall\, i=1,\dots,m$

Can define a vector fn $h:\mathbb{R}^n \longrightarrow \mathbb{R}^m$ with $h_1,\dots,h_m$ as components

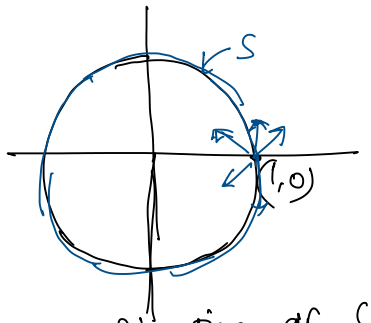
$h(x)=0$

Equality constraints: $f:\mathbb{R}^n \to \mathbb{R}, h:\mathbb{R}^n \to \mathbb{R}^m$

$$\min_{x\in\mathbb{R}^n} f(x) \quad \text{s.t. } h(x)=0\ (h_i(x)=0, i=1,\dots,m) \equiv \min_{x\in S} f(x), \quad S=\{x\in\mathbb{R}^n : h(x)=0\}$$

feasible directions

$h(x)=x_1^2+x_2^2-1$



feasible directions at (1,0)

For d to be feasible

$\begin{bmatrix}1\\0\end{bmatrix}+\alpha\begin{bmatrix}d_1\\d_2\end{bmatrix}\in S$ for small α $\forall \alpha\in(0,\epsilon)$

$(1+\alpha d_1)^2+(\alpha d_2)^2-1=0$

$\cancel{1}+\alpha^2 d_1^2+2\alpha d_1+\alpha^2 d_2^2=\cancel{1}$

>0

$d_2=0$

(cannot be satisfied) $\alpha^2 d_1^2+2\alpha d_1=0$

$\Rightarrow$ there is no feasible directions

first order feasible direction

d is called a 1st order feasible direction if

$h(x+\alpha d)=o(\alpha)$ ✓

$\Leftrightarrow \dfrac{h(x+\alpha d)}{\alpha} \xrightarrow{\alpha\to 0} 0$

$h(x+\alpha d)$ goes to 0 faster than α goes to 0

Eg, $h(x+\alpha d)=\alpha^2$

$\dfrac{h(x+\alpha d)}{\alpha}=\alpha \longrightarrow 0$

$$\lim_{\alpha\to 0}\frac{h(x+\alpha d)}{\alpha}=0$$

$$\Rightarrow \lim_{\alpha\to 0}\frac{h(x+\alpha d)-h(x)}{\alpha}=0$$

$\Rightarrow \boxed{\nabla h(x)^T d = 0}$ ✓

directional derivative of h at x in direction d

If d is a feasible direction $\Rightarrow$ 1st order feasible direction

Suppose d is feasible direction at x

$h(\underbrace{x+\alpha d}_{\in S}) = 0$ for all small α

But 1st order feasible $\not\Rightarrow$ feasible

Next time

$S = \{x \in \mathbb{R}^n : h(x) = 0\}$

d is called feasible dir^n at x if $x + \alpha d \in S$ $\forall$ small enough α.

$\Leftrightarrow h(x+\alpha d) = 0$ ✓

d is called first order feasible dir^n at x if $h(x+\alpha d) = o(\alpha)$

$\left(\lim_{\alpha \to 0} \frac{h_i(x+\alpha d)}{\alpha} = 0\right) \forall i$ $\quad \lim_{\alpha \to 0} \frac{h(x+\alpha d)}{\alpha} = 0$

$\Leftrightarrow \nabla h_i(x)^T d = 0 \quad \forall i$

$$\nabla h(x) = \underbrace{\begin{bmatrix} \nabla h_1(x), & \cdots, & \nabla h_m(x) \end{bmatrix}}_{n \times m} \Rightarrow \nabla h(x)^T = \begin{bmatrix} \text{---} \nabla h_1(x)^T \text{---} \\ \text{---} \nabla h_2(x)^T \text{---} \\ \\ \text{---} \nabla h_m(x)^T \text{---} \end{bmatrix} d$$

$\Leftrightarrow \nabla h(x)^T d = 0$

First order feasible directions: Given a $x \in \mathbb{R}^n$, $d \in \mathbb{R}^n$ is called a first order feasible direction at x if $\nabla h(x)^T d = 0$ ($\nabla h_i(x)^T d = 0, i = 1, \cdots, m$).

- Each feasible direction is a first order feasible direction but each first order feasible direction need not be feasible.

Set of first order feasible directions at x

$$V(x) = \left\{d \in \mathbb{R}^n : \nabla h(x)^T d = 0\right\}$$

- This null space of $\nabla h(x)^T$.

Regular points: A $x \in \mathbb{R}^n$ is called a regular point if

1. $h(x) = 0$ (x is feasible)
2. $\nabla h_i(x), i = 1, \cdots, m$ are linearly independent.

Lagrange Multiplier Theorem (necessary conditions for local optimality): Let f and h be continuously differentiable. Let x^* be a local minimum that is also regular. Then there exists a unique $\lambda^* \in \mathbb{R}^m$, called Lagrange multiplier, such that

$$\nabla f(x) + \nabla h(x)\lambda^* = 0$$

If in addition f and h are twice continuously differentiable then

$$d^T\left(\nabla^2 f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla^2 h_i(x^*)\right)d \geq 0 \text{ for all } d \in V(x^*).$$

- These are also sufficient conditions if f is convex and the constraint set $S = \{x \in \mathbb{R}^n : h(x) = 0\}$ is also convex.

Approach to obtain x^*: (a) Solve

$$h(x) = 0$$
$$\nabla f(x) + \nabla h(x)\lambda = 0$$

for $n + m$ unknowns x and λ.
(b) Impose second order necessary conditions if applicable.
(c) Inquire each of the candidate solutions.

Lagrange function: $L: \mathbb{R}^{n \times m} \to \mathbb{R}$

$$L(x, \lambda) = f(x) + \lambda^T h(x)$$

Necessary conditions: Let x^* be a local minimum that is also regular. Then there exists a unique $\lambda^* \in \mathbb{R}^m$

$$\nabla_x L(x^*, \lambda^*) = 0$$

$d^T \nabla^2_{xx} L\left(x^*, \lambda^*\right) d \geq 0$ for all $d \in V(x^*)$

Sufficient conditions for local optimality: Let f and h are twice continuously differentiable and x^* and λ^* be such that

$$h(x^*) = 0$$
$$\nabla_x L\left(x^*, \lambda^*\right) = 0$$
$$d^T \nabla^2_{xx} L\left(x^*, \lambda^*\right) d > 0 \text{ for all } d \in V(x^*) \setminus \{0\}$$

Then x^* is a strict local minimum.

Sensitivity: Let $x^*(u)$ be an optimal solution and $p(u) \triangleq f(x^*(u))$ be the optimal value. The rate of change of the optimal value with respect to u, $\nabla p(u)$, is referred to as sensitivity.

Sensitivity Theorem: Let $x^*(u)$ be regular and $\left(x^*(u), \lambda^*(u)\right)$ be an optimal solution – Lagrange multiplier pair that satisfies the sufficient conditions then

$$\nabla p(u) = -\lambda^*(u)$$